

Environmental Management

Exam Preparation — Complete Answers

Based on Environmental Management Book & ISO 14000/EMS Source

Q1. Differentiate Between Food Chain and Food Web

Food Chain

A food chain is the sequence of **eating and being eaten** in an ecosystem. It is a linear, unidirectional pathway through which **energy and nutrients** flow from one organism to another. All organisms — living or dead — are potential food for some other organism.

Types of food chains:

- **Grazing Food Chain:** Begins with **green plants** (primary producers) and ends with carnivores. Example: Grass → Grasshopper → Frog → Snake → Hawk
- **Detritus Food Chain:** Begins with dead organic matter consumed by **detritivores** and decomposers. Example: Dead leaves → Fungi → Bacteria (Forest ecosystem)

Food Web

Food chains in ecosystems are **rarely isolated**. They are interconnected with several **linkages** and form a **complex network** known as a food web. A food web has multiple options of eating and being eaten at each trophic level.

Differences

Food Chain: Linear and single-pathway. One organism eats only one specific organism at each level.

Food Web: Network of interconnected food chains. Multiple organisms at each trophic level can eat or be eaten by many others.

Food Chain: Less stable — if one species is removed, the entire chain collapses.

Food Web: More stable — if one species is removed, alternatives exist at each trophic level.

Food Chain: Simpler and theoretical.

Food Web: More realistic representation of nature.

Q2. What is an Ecological Pyramid?

An ecological pyramid is a graphic representation of the trophic structure and function of an ecosystem, starting with producers at the base and successive trophic levels forming the apex. There are three types:

1. Pyramid of Numbers

Represents the number of **individual organisms** at each trophic level. It can be upright (grassland, pond) or inverted (parasitic food chain). In a grassland, producers (grasses) are most numerous; in a parasitic chain, one tree supports many birds and even more parasites — producing an inverted pyramid.

2. Pyramid of Biomass

Based on the total dry biomass (weight of living matter) at each trophic level. In a forest, it is upright — trees accumulate huge biomass while consumer biomass decreases upward. In a pond, it is inverted — phytoplankton biomass is less than that of zooplankton and fish above them.

3. Pyramid of Energy

Represents the amount of energy at each trophic level. The pyramid of energy is always upright because at each successive trophic level, approximately 90% of the energy is lost as heat and respiration. Only about 10% of energy passes to the next level. It provides the best representation of trophic relationships.

Q3. What Are the Functions and Powers of CPCB?

The Central Pollution Control Board (CPCB) is the main regulatory body for pollution control in India, constituted under the Water (Prevention and Control of Pollution) Act, 1974. Its functions and powers include:

- Advises the Central Government in matters related to prevention and control of water and air pollution.
- Coordinates the activities of State Pollution Control Boards and provides them technical assistance and guidance.
- Organizes training programs for prevention and control of pollution.
- Organizes comprehensive programs on pollution-related issues through mass media.
- Collects, compiles and publishes technical and statistical data related to pollution.
- Prepares manuals for treatment and disposal of sewage and trade effluents.
- Lays down standards for water quality parameters.
- Plans nation-wide programs for prevention, control or abatement of pollution.
- Establishes and recognizes laboratories for analysis of water, sewage or trade effluent samples.

State Pollution Control Boards operate under the direction of CPCB at the state level and perform similar functions including advising state governments on industry locations, laying down effluent standards, granting or refusing consent to industries after effluent analysis.

Q4. Explain TQM (Total Quality Management)

Total Quality Management (TQM) is a management philosophy and approach that focuses on continuous improvement of processes, products, and services by involving all members of an organization. It aims to meet or exceed customer expectations consistently.

Definition

TQM is a customer-focused approach that involves all employees in the ongoing improvement of an organization's processes and outputs. It integrates quality into every activity of the organization.

Key Principles of TQM

1. Customer Focus: Understanding and meeting the needs of customers.
2. Total Employee Involvement: All employees participate in working toward common goals.
3. Process-Centered: A fundamental part of TQM is a focus on process thinking.
4. Integrated System: Connecting the horizontal processes, horizontal organizational connections, and the need to understand functions.
5. Strategic & Systematic Approach: A strategic plan to achieve the quality vision.
6. Continual Improvement: TQM places heavy emphasis on continuous improvements of processes.
7. Fact-Based Decision Making: Making decisions based on data collected through performance metrics.
8. Communication: Changes in strategy, policy, and execution must be communicated effectively.

Benefits of TQM

- Improved product and service quality
- Increased customer satisfaction
- Reduction in waste and defects

- Higher employee morale and involvement
 - Better competitiveness and business performance
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Q5. Explain the Ishikawa Diagram

The Ishikawa Diagram, also known as the **Cause-and-Effect** Diagram or Fishbone Diagram, is a TQM tool developed by Kaoru Ishikawa. It is used to identify and analyze the **root** causes of a problem or defect in a process.

Structure

The diagram resembles the skeleton of a fish. The problem or effect is placed at the 'head' (right side), and the possible causes are represented as 'bones' branching off the spine. The main categories of causes are often grouped as:

- Man (People): Human errors, skill gaps, lack of training
- Machine (Equipment): Equipment failure, improper tools
- Method (Process): Incorrect procedures, poor workflow
- Material: Defective inputs or raw materials
- Measurement: Inaccurate data collection or measuring tools
- Environment: Working conditions, temperature, noise

Uses

- Root cause analysis of quality problems
- Identifying contributing factors in a structured way
- Brainstorming and organizing team ideas
- Supporting continuous improvement in TQM

How to Draw It

9. Write the problem/effect at the right end (fish head).
 10. Draw a horizontal arrow pointing to it (the spine).
 11. Draw diagonal branches for each main cause category.
 12. Add sub-causes as smaller branches.
 13. Analyze the diagram to identify the most likely root causes.
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Q6. What is ISO 14000?

ISO stands for the International Organisation for Standardization, located in Geneva, Switzerland. ISO 14000 refers to a family of voluntary standards and guidance documents designed to help organizations address environmental issues.

Key Components of the ISO 14000 Family

- Standards for Environmental Management Systems (EMS)
- Environmental and EMS auditing standards
- Environmental labeling standards
- Environmental performance evaluation standards
- Life-cycle assessment standards

ISO 14001 — The Foundation Document

ISO 14001, first published in September 1996, is the most important standard in the ISO 14000 family. It is an international voluntary standard describing specific requirements for an EMS. It is the dominant Environmental Management System standard in the world. A second

edition was published in 2004, and it was further updated in 2015. An organization can receive certification or registration to ISO 14001.

Purpose of ISO 14000

- Reduce environmental impacts of organizations
 - Provide a voluntary framework for environmental responsibility
 - Assist organizations in complying with environmental regulations
 - Improve overall operating efficiency and demonstrate commitment to sustainability
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Q7. Explain the Stages of ISO 14001 (Stages of EMS)

The implementation of ISO 14001 follows a cycle of five stages, revolving around the principle of continuous improvement:

Stage 1: Commitment and Policy

The organization's top management establishes a clear environmental policy that defines its commitment to the environment. This policy serves as the foundation of the entire EMS and must be regularly evaluated and updated to reflect changing environmental priorities.

Stage 2: Planning

In this phase, the organization identifies its environmental aspects (elements that can interact with the environment) and impacts (actual changes to the environment). It then sets measurable objectives and targets to minimize these impacts and establishes action plans to meet them. It also ensures compliance with applicable environmental laws.

Stage 3: Implementation and Control

This stage is carried out in two steps:

14. The organization collects all its resources and follows the action plan to achieve the set objectives. Training and awareness are provided to all employees.
15. Documentation is completed and both internal and external communication channels are established. Roles, responsibilities, and authorities are clearly defined.

Stage 4: Evaluation (Check)

The current execution steps are evaluated to analyze performance and determine whether the required objectives have been achieved. Key characteristics of operations are monitored and measured. If objectives are not met, corrective measures are taken.

Stage 5: Review and Improvement (Act)

Regular reviews by top management ensure the EMS continues to meet organizational needs. The final computed results are analyzed to verify whether the EMS is working effectively. This stage creates a loop of continuous improvement for the organization.

Q8. What is EMS? Explain the PDCA Cycle.

Environmental Management System (EMS)

An Environmental Management System (EMS) is a set of **processes and practices** that enable an organization to reduce its **environmental impacts** and increase its operating efficiency. It includes the organizational structure, planning and resources for developing, implementing and maintaining policy for environmental protection.

The aims of EMS are:

- To increase compliance with environmental regulations

- To reduce waste and minimize environmental impact

PDCA Cycle (Plan-Do-Check-Act)

The EMS is structured around the PDCA cycle — an iterative four-stage approach for continuous improvement:

PLAN: Identify environmental aspects (e.g. air pollutants), determine the most significant ones, understand legal requirements, and set objectives and targets (e.g. reduce emissions by 10% by a specific date). Develop management action plans.

DO: Implement the plan. This includes defining the management structure with roles and responsibilities, providing training and awareness to employees, establishing communication procedures (internal and external), maintaining EMS documentation, ensuring document control, operational control, and emergency preparedness and response.

CHECK: Monitor and measure key characteristics of operations. Establish corrective action procedures for non-conformance. Conduct periodic EMS audits to verify whether the EMS conforms to ISO 14000 standards and has been properly implemented.

ACT: Top management reviews the EMS to ensure its continuing suitability, adequacy, and effectiveness. Evaluate the EMS for suitability of operation and consider continual improvement as part of operations. This final review creates a loop back to the planning stage.

Q9. What Are the Different TQM Tools?

TQM uses several analytical and problem-solving tools to monitor and improve quality. The major TQM tools are:

16. Ishikawa (Fishbone/Cause-and-Effect) Diagram: Identifies root causes of a problem.
17. Pareto Chart: A bar graph showing the frequency of problems or causes, helping prioritize the most significant ones (80-20 rule).
18. Control Charts: Statistical charts used to monitor process variation over time and determine if the process is in control.
19. Flowcharts: Diagrams representing the sequence of steps in a process, useful for identifying inefficiencies.
20. Histograms: Bar graphs showing the distribution of data — used to understand variation in a process.
21. Scatter Diagrams: Plots two variables to identify potential relationships or correlations between them.
22. Check Sheets: Simple data collection forms for systematically recording and compiling data.

These seven tools are often referred to as the '7 QC Tools' (Seven Quality Control Tools) and form the backbone of TQM problem-solving.

Q10. Explain Any TWO TQM Tools in Detail

Tool 1: Pareto Chart

A Pareto Chart is a bar graph used in quality management to identify and prioritize the most significant problems or causes contributing to a quality issue. It is based on the Pareto Principle (80-20 Rule), which states that approximately 80% of problems are caused by 20% of causes.

Structure: The x-axis shows categories of problems/causes arranged in descending order of frequency. The y-axis shows frequency or impact. A cumulative percentage line is overlaid.

Uses:

- Helps focus improvement efforts on the most impactful causes
- Prioritizes problems based on data
- Used in root cause analysis and continuous improvement

Example: In a manufacturing unit, defects may occur due to machine failure, human error, material defect, or poor method. A Pareto Chart reveals that 80% of defects are caused by machine failure and human error — directing attention to these two causes first.

Tool 2: Control Chart

A Control Chart (also called a Shewhart Chart) is a statistical process control tool used to monitor whether a process remains stable and within control over time. It shows data plotted in time order with control limits.

Components:

- Centre Line (CL): The mean or average of the process
- Upper Control Limit (UCL): Set at +3 standard deviations from the mean
- Lower Control Limit (LCL): Set at -3 standard deviations from the mean

Uses:

- Identifies when a process is going out of control before defects occur
- Distinguishes between common cause (random) variation and special cause (assignable) variation
- Helps maintain consistent quality over time

Example: A control chart tracking daily production defect rates will alert managers if defect rates exceed the UCL, triggering investigation and corrective action.

Q11. Define CSR and Give Any Two Examples in India

Definition of CSR

Corporate Social Responsibility (CSR) is a **business model** by which companies integrate **social, environmental, and ethical concerns** into their **business operations** and interactions with stakeholders. It is the responsibility of a corporation toward society and the environment **beyond just making profit**.

Under the Companies Act, 2013 in India, companies with a net worth of Rs. 500 crore or more, or turnover of Rs. 1000 crore or more, or net profit of Rs. 5 crore or more, are required to spend at least 2% of their average net profit on CSR activities.

CSR Helps In:

- Energy efficiency
- Emission reduction
- Biodiversity conservation
- Community development

Examples of CSR in India

1. Tata Group: Tata companies have long-standing CSR initiatives including Tata Trusts — involved in education, healthcare, rural development, and drinking water projects across India. Tata Steel runs community development programs in Jharkhand and Odisha.

2. Infosys Foundation: Infosys has funded the construction of hospitals, schools, and libraries across India. It has also supported rehabilitation of flood-affected villages and rural infrastructure development.

Q12. What Are the Benefits of CSR?

The benefits of Corporate Social Responsibility (CSR) can be seen at multiple levels — for the company, employees, society, and the environment:

For the Company

- Improved company reputation and brand value among customers and stakeholders
- Competitive and financial advantage through improved efficiencies and reduced costs
- Stronger stakeholder trust and investor confidence
- Easier compliance with regulations and legal requirements

For Employees

- Improved employee morale and job satisfaction
- Increased engagement and sense of purpose
- Attracts talent who value ethical organizations

For Society and Environment

- Contributes to energy efficiency and emission reduction
- Supports biodiversity conservation
- Promotes education, healthcare, and community well-being
- Encourages sustainable practices in supply chains

Other Benefits

- Drives sustained environmental performance
- Encourages better environmental performance of suppliers
- A focus on life-cycle thinking — considering environment from development to end-of-life

Q13. Write a Short Note on CER (Corporate Environmental Responsibility)

Corporate Environmental Responsibility (CER) is a **component of CSR** that specifically focuses on an **organization's responsibility** toward the natural environment. It refers to the obligations of a corporation to minimize its negative environmental impacts and contribute to environmental sustainability.

Key Areas of CER

- Reducing **greenhouse** gas emissions and carbon footprint
- **Energy** efficiency — using renewable energy sources
- **Waste** reduction and responsible disposal of hazardous materials
- **Water** conservation and preventing water pollution
- Biodiversity conservation and protecting natural ecosystems
- Compliance with environmental laws and regulations (e.g. ISO 14001, EPA 1986)

Importance of CER

As industries grow, their environmental footprint increases. CER ensures that businesses take proactive steps to offset their environmental impact. It also helps companies avoid legal penalties, improve public image, and achieve long-term sustainability.

Relationship with CSR

CER is a subset of CSR. While CSR encompasses social, ethical, and economic responsibilities, CER is specifically concerned with environmental obligations. Companies that adopt EMS frameworks like ISO 14001 are effectively practicing CER.

Q14. Write a Short Note on the Environment (Protection) Act, 1986

The Environment (Protection) Act, 1986 came into force on **November 19, 1986** — the birth anniversary of Late Prime Minister **Indira Gandhi**, who was a pioneer of environmental protection in India. The Act extends to the whole of India and is often referred to as an 'umbrella Act' as it encompasses earlier environmental legislation.

Key Definitions under the Act

- **Environment**: Includes water, air, land and the inter-relationships that exist among them and human beings, all living organisms, and property.
- **Environmental Pollution**: The presence of any solid, liquid, or gaseous substance in concentration that may be injurious to the environment.
- **Hazardous Substance**: Any substance liable to cause harm to human beings, living organisms, property, or the environment.

Powers of Central Government under the Act

- Setting standards for quality of air, water, and soil for various areas.
- Specifying maximum **permissible** limits of environmental pollutants (including noise).
- Laying down procedures for handling hazardous substances.
- Imposing **prohibition** and **restrictions** on industries in certain areas.
- Providing procedures and safeguards for prevention of environmental accidents.

Other Important Provisions

- Environmental Impact Assessment (EIA) under Schedule I — 29 types of projects require Central Government clearance.
 - Hazardous Wastes (Management and Handling) Rules, 1989 — 18 categories of hazardous waste are recognized.
 - Environmental Audit provisions to check whether companies comply with environmental laws.
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Q15. What is the Wildlife (Protection) Act, 1972?

The Wildlife (Protection) Act, 1972 is a landmark legislation in the history of wildlife conservation in India. It came into existence in 1972, and wildlife was subsequently transferred from the State list to the Concurrent list in 1976, giving the Central Government power to enact comprehensive legislation.

Major Provisions of the Act

- Defines wildlife-related terminology.
- Provides for the appointment of a Wildlife Advisory Board, Wildlife Warden, and defines their powers and duties.
- Provides comprehensive listing of endangered wildlife species and imposes prohibition on hunting of endangered species.
- Provides protection to some endangered plants like Beddome's Cycad, Blue Vanda, Ladies Slipper Orchid, and Pitcher Plant.
- Provides for setting up of National Parks and Wildlife Sanctuaries.

- Provides for the constitution of the Central Zoo Authority.
- Imposes a ban on trade or commerce in scheduled animals.
- Provides legal powers to officers and specifies punishment for offenders.
- Provides for captive breeding programmes for endangered species.

Conservation Projects under the Act

- Project Lion (1972)
- Project Tiger (1973)
- Crocodile Conservation (1974)
- Brown Antlered Deer Conservation (1981)

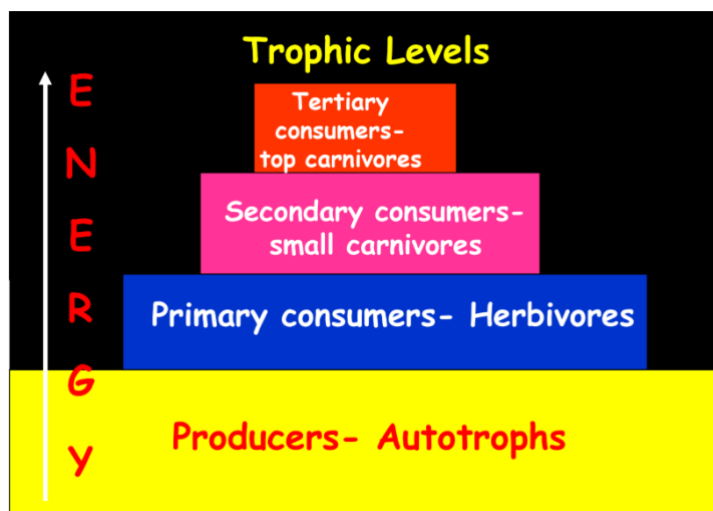
Drawbacks

The Act has been criticized for mild penalties to offenders, illegal wildlife trade in J&K (which has its own separate act), and little emphasis on plant genetic resource protection.

Q16. Draw an Ecological Pyramid

An ecological pyramid is a diagrammatic representation of the trophic structure of an ecosystem, with producers at the base and top carnivores at the apex.

Pyramid of Energy (always upright)



Top Carnivores	(e.g. Hawk, Eagle)
Carnivores	(e.g. Snake, Frog)
Herbivores	(e.g. Grasshopper, Rabbit)
Producers	(e.g. Grasses, Plants) [BROADEST BASE]

Types of Ecological Pyramids:

- Pyramid of Numbers: Represents number of organisms at each level. Can be upright (grassland) or inverted (parasitic chain).
- Pyramid of Biomass: Represents total dry weight at each trophic level. Upright in forests, inverted in aquatic ecosystems.
- Pyramid of Energy: Represents energy at each trophic level. Always upright. At each level, 90% of energy is lost, only 10% passes up.

Q17. Explain the PDCA Cycle

The PDCA (Plan-Do-Check-Act) Cycle, also known as the Deming Cycle, is an iterative four-stage management model used for continuous improvement in processes, products, or services. It is the foundation of both ISO 14001 (EMS) and TQM frameworks.

PLAN

Identify the problem or opportunity for improvement. Set objectives and targets. Identify environmental aspects and their impacts. Understand applicable legal and regulatory requirements. Develop a management action plan to achieve objectives. Example: Objective — reduce CO2 emissions; Target — reduce by 10% within 12 months.

DO

Implement the plan. Assign roles, responsibilities, and authorities. Provide training and awareness to all employees so they understand their environmental roles. Establish communication channels (internal and external). Maintain documentation and document control. Ensure operational control and emergency preparedness.

CHECK

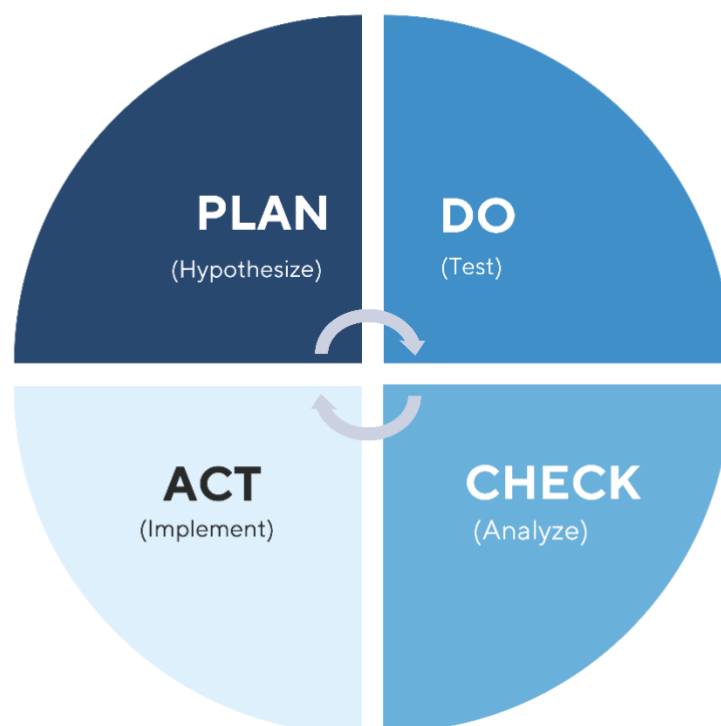
Monitor and measure results against the objectives and targets. Evaluate whether the plan has been successfully implemented. Conduct EMS audits. Identify non-conformances and establish corrective actions. Analyze performance data to assess effectiveness.

ACT

Top management reviews the overall EMS or process system. Evaluate continuing suitability and effectiveness. Take corrective and preventive actions based on findings. Make permanent improvements. Feed lessons learned back into the PLAN stage to begin a new cycle of improvement.

The PDCA Cycle ensures that improvements are systematic, data-driven, and continuous — making it a central concept in both environmental management and quality management.

The PDCA Cycle



Q18. What Are the Objectives of Six Sigma?

Six Sigma is a data-driven quality management methodology that aims to improve processes by identifying and eliminating defects and minimizing variability. The term 'Six Sigma' refers to a statistical measure where a process produces no more than 3.4 defects per million opportunities (DPMO).

Main Objectives of Six Sigma

23. Reduce Defects: Minimize the number of defects and errors in products or processes to near-zero levels.
24. Reduce Process Variation: Eliminate variability in processes to ensure consistent, predictable outputs.
25. Improve Quality: Enhance the overall quality of products and services to meet or exceed customer expectations.
26. Increase Customer Satisfaction: Deliver consistent quality, thereby improving customer loyalty and satisfaction.
27. Reduce Costs: Fewer defects and reduced rework lead to significant cost savings.
28. Improve Process Efficiency: Streamline and optimize processes to eliminate waste (time, resources, effort).
29. Data-Driven Decision Making: Encourage decisions based on measurement and statistical analysis rather than assumptions.
30. Achieve Breakthrough Improvement: Target transformative improvements (not just incremental) in key business processes.

Six Sigma Methodologies

DMAIC: Define, Measure, Analyze, Improve, Control — used for improving existing processes.

DMADV: Define, Measure, Analyze, Design, Verify — used for creating new processes or products.

Levels (Belts) in Six Sigma

- White Belt, Yellow Belt, Green Belt, Black Belt, Master Black Belt — indicating increasing levels of expertise and leadership in Six Sigma projects.